

Module 02

1. Fraud Transaction Detection (Linear Search)

÷ C++ Code

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter the value of n:";
    cin >> n;

    int arr[n];
    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
    }

    int target;
    cin >> target;
    bool found = false;
    for (int i = 0; i < n; i++)
    {
        if (arr[i] == target)
        {
            found = true;
            break;
        }
    }

    if (found)
        cout << "Fraud Transaction found";
    else
        cout << "Transaction not found";
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\sjha0\Desktop\cpp\2nd_week_module> cd "c:\Users\sjha0\Desktop\cpp\2nd_week_module\" ; if ($?) { g++ fraud.cpp -o fraud } ; if ($?) { .\fraud }
```

enter the value of N:4

2342 4342 63534 4224

enter the value of target: 4342

Fraud Transaction Found

```
PS C:\Users\sjha0\Desktop\cpp\2nd_week_module> █
```

2. Search Log Search (Binary Search)

C++

```
#include <iostream>
using namespace std;
int main() {
    int N;
    cin >> N;
    int arr[N];
    for (int i = 0; i < N; i++) {
        cin >> arr[i];
    }
    int target;
    cin >> target;
    int low = 0, high = N - 1;
    bool found = false;
    while (low <= high) {
        int mid = (low + high) / 2;
        if (arr[mid] == target) {
            found = true;
            break;
        }
        else if (arr[mid] < target)
            low = mid + 1;
        else high = mid - 1;
    }
    if (found)
        cout << "Log found";
    else
        cout << "Log not found";
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\sjha\Desktop\cpp\2nd_week_module> cd "c:\Users\sjha\Desktop\cpp\2nd_week_module\" ; if ($?) { g++ serve_log_search.cpp -o serve_log_search } ; if ($?) { .\serve_log_search }
enter the value of N: 4
3456 7685 5756 6789
3456
Log Found
PS C:\Users\sjha\Desktop\cpp\2nd_week_module> |
```

3. Sort Employee Stories (Bubble Sort)

C++

```
#include <iostream>
using namespace std;
int main() {
    int N;
    cout << "Enter the value of N: ";
    cin >> N;
    int arr[N];
    for (int i = 0; i < N; i++) {
        cin >> arr[i];
    }
    for (int i = 0; i < N - 1; i++)
        for (int j = 0; j < N - i - 1; j++)
            if (arr[j] > arr[j + 1])
            {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
    for (int i = 0; i < N; i++)
        cout << arr[i] << " ";
    return 0;
}
```

```
PS C:\Users\sjha0\Desktop\cpp\2nd_week_module> cd "c:\Users\sjha0\Desktop\cpp\2nd_week_module\" ; if ($?) { g++ employee_salaries.cpp -o employee_salaries } ; if ($?) { .\employee_salaries }  
enter the value of N: 5  
4525235 486885 525468 968964 523555  
486885 523555 525468 968964 4525235  
PS C:\Users\sjha0\Desktop\cpp\2nd_week_module> |
```


4. Sort Website Response Time (Insertion Sort)
→ C++

```
#include <iostream>
using namespace std;
int main() {
    int N;
    cout << "Enter N: ";
    cin >> N;
    int arr[N];
    for (int i = 0; i < N; i++)
        { cin >> arr[i]; }
    for (int i = 1; i < N; i++)
        { int key = arr[i];
          int j = i - 1;
          while (j >= 0 && arr[j] > key)
              { arr[j+1] = arr[j];
                j--;
              }
          arr[j+1] = key;
        }
    for (int i = 0; i < N; i++)
        { cout << arr[i] << " "; }
    return 0;
}
```

```
PS C:\Users\sjha0\Desktop\cpp\2nd_week_module> cd "c:\Users\sjha0\Desktop\cpp\2nd_week_module\" ; if ($?) { g++ sort_website.cpp -o sort_website } ; if ($?) { .\sort_website }  
enter the value of N: 4  
23 54 66 34  
23 34 54 66  
PS C:\Users\sjha0\Desktop\cpp\2nd_week_module> |
```


5. Prioritize Bug Report (selection sort)

÷ C++

```
#include <iostream>
using namespace std;
int main () {
    int N;
    cout << "Enter N ";
    cin >> N;
    int arr[N];
    for (int i=0; i<N; i++) {
        cin >> arr[i];
    }
    for (int i=0; i<N-1; i++) {
        int minIndex = i;
        for (int j=i+1; j<N; j++) {
            if (arr[j] < arr[minIndex]) {
                minIndex = j;
            }
        }
        int temp = arr[i];
        arr[i] = arr[minIndex];
        arr[minIndex] = temp;
    }
    for (int i=0; i<N; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\sjha0\Desktop\cpp\2nd_week_module> cd "c:\Users\sjha0\Desktop\cpp\2nd_week_module\" ; if ($?) { g++ bug.cpp -o bug } ; if ($?) { .\bug }
```

```
enter the value of N: 6
```

```
1 4 3 9 2 7
```

```
1 2 3 4 7 9
```

```
PS C:\Users\sjha0\Desktop\cpp\2nd_week_module> █
```